

# Explaining Conformal Prediction: Diagnosing Reliability through Feature-Conditioned Outcomes and p-Value Margins

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## Abstract

Conformal prediction produces prediction sets with finite-sample coverage guarantees, and its practical utility for decision-making grows when the reliability of its predictions can be explained at the feature level. Yet, the intersection of conformal prediction and explainability remains relatively underexplored, leaving practitioners without tools to interpret how input features shape prediction reliability.

We introduce a framework for explaining the reliability of conformal classifiers at the feature level using two tools. First, Feature-Conditioned Outcome Distribution (FCOD) plots visualize how correct, incorrect, and ambiguous conformal prediction set outcomes vary across feature ranges, identifying regions of the feature space associated with different empirical behaviors in terms of correctness and uncertainty of the predictions. Second, we propose two metrics derived from class-conditional conformal p-values: the label-free evidence margin, which provides a signed contrast between competing classes without the true label, and the label-dependent correctness margin, which compares the conformal p-value of the true class with that of its strongest alternative. They are used as SHAP targets for local feature attribution and aggregated global analysis, enabling feature-level explanations of relative conformal evidence between competing classes and support for the true class.

We illustrate the framework on a synthetic multiclass dataset and on a real binary classification problem, the Hotel Booking Demand dataset, showing that FCOD plots and margin-based SHAP explanations provide complementary diagnostic insights: FCOD plots identify where in the feature space reliable and unreliable behavior concentrates, while SHAP analyses of the p-value margins identify how the features contribute to the model calibration, supporting more informed decision-making in real-world classification settings.

**Keywords:** Conformal prediction, Explainability, Uncertainty quantification, SHAP, Shapley values

## 1. Introduction

The increasing use of machine learning models in decision-making tasks across diverse application domains has highlighted the importance of explainability and robust uncertainty quantification of their outputs. In this context, conformal prediction provides a framework for uncertainty quantification with finite-sample coverage guarantees under minimal distributional assumptions that are applicable to any machine learning model (Vovk et al., 2005; Shafer and Vovk, 2008). More precisely, in classification tasks, conformal classifiers return prediction sets that guarantee to contain the true label with a probability of at least  $1 - \alpha$ , where  $\alpha$  is the significance level determined by the user. This distribution-free validity has driven growing adoption in high-stakes applications (Angelopoulos and Bates, 2023; Fontana et al., 2023).

Conformal prediction provides statistically calibrated uncertainty estimates, improving the information available to decision-makers in real-world settings. Its practical value increases when the conformal model is endowed with explainability, offering insight into how input features shape prediction reliability. Yet, despite its considerable potential, the combination of conformal prediction and explainability has only been explored to a limited extent. A remarkable recent work along these lines was the proposal by Johansson et al. (2025), which pioneered the integration of conformal p-values, used to assess whether the hypothesis that an instance belongs to a particular class can be rejected at a given significance level, with a popular explainability tool, namely SHAP values (Lundberg and Lee, 2017), for explaining prediction sets produced by conformal classifiers. They acknowledge the potential of SHAP values for explaining different properties of conformal classification outputs, and our work further advances this direction.

We propose a framework that equips a conformal classifiers with an explainability layer that exposes the feature-level drivers of prediction reliability. It is based on conformal p-values, which offer a measure of plausibility, quantifying how well a new instance conforms to a given class.

As a first explainability layer, we propose using a feature-wise visualization of the conditional distribution of outcome categories of a conformal classifier. These categories may consist of a single correct label, a single incorrect label, or multiple labels. The resulting plots characterize how the distribution of conformal prediction outcome categories varies across the values of a given feature. This enables relating feature values to both the classifier’s confidence and its propensity to make correct or incorrect predictions. We call them Feature-Conditioned Outcome Distribution plot (FCOD plot).

Our second explainability layer is based on a set of metrics derived from conformal p-values. These measures aggregate the p-values obtained for each instance and class to quantify both the classifier’s confidence in its decision relative to competing classes and whether this confidence ultimately corresponds to a correct prediction. Thus, we propose a label-dependent metric, which requires access to ground-truth labels, and a label-free one, which relies solely on model outputs. In particular, for a given instance, we propose the following metrics:

- Evidence margin, defined as the difference between the p-values of two specific classes. This measure quantifies how strongly the classifier separates the two classes, independently of the true label.

- Correctness margin, defined as the difference between the p-value of the true class and that of the most competitive alternative. It links the classifier’s certainty and its propensity to produce correct or incorrect predictions.

We then use these metrics as SHAP targets to provide instance-level attributions of predictive outcomes and their associated reliability, in contrast to the global, feature-wise patterns captured by FCOD plots. In this way, the metrics are operationalized into a tool that can guide practitioners in diagnosing how specific feature values relate to the performance of the conformal classifier. The Python code needed to run these tools is available on GitHub as a library <sup>1</sup>.

We illustrate our tools using two datasets: a synthetic multiclass dataset, which highlights the multiclass capabilities of our approach, and the Hotel Booking Demand dataset (Antonio et al., 2019), where the task is to predict whether a reservation will be canceled, providing a simple and intuitive real-world setting. Our results demonstrate that the proposed tools expose both the features and the specific feature values that govern prediction reliability, offering practitioners concrete diagnostic signals that support trustworthy decision-making.

The rest of the paper is organized as follows: Section 2 reviews related work; Section 3 defines FCOD plots and our margin objectives; Section 4 describes the experimental setup; Section 5 presents our findings; and Section 6 concludes.

## 2. Related work

Conformal prediction (Vovk et al., 2005; Shafer and Vovk, 2008) provides a framework for constructing prediction sets with finite-sample coverage guarantees under minimal distributional assumptions. A key development for practical applications is the inductive (split) conformal approach (Papadopoulos et al., 2002; Toccaceli, 2022), which separates training and calibration data, enabling computational efficiency while preserving validity. The framework has been extended in several directions: Romano et al. (2020) developed adaptive conformal methods for classification that achieve valid coverage under complex data distributions, Lei et al. (2018) developed distribution-free predictive inference for regression, and Bates et al. (2021) broadened the applicability of conformal ideas to risk-controlling prediction sets.

Mondrian conformal prediction (Vovk et al., 2005; Boström et al., 2021) partitions the calibration set by class label to ensure class-conditional coverage. This is particularly valuable in imbalanced datasets or when per-class reliability is required.

### 2.1. Explainability and Conformal Prediction

SHAP (Lundberg and Lee, 2017) interprets machine learning predictions by assigning each feature a contribution value derived from Shapley values (Shapley, 1953), a concept from cooperative game theory. These values decompose a prediction into additive feature contributions while satisfying local accuracy, missingness, and consistency, providing a theoretically grounded explanation framework. Although exact Shapley value computation scales

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1. <https://github.com/Caparrini/conformalpy>

exponentially with the number of features, efficient approximations exist. In particular, KernelSHAP (Lundberg and Lee, 2017) makes SHAP practical by providing a model-agnostic approximation based on weighted linear regression. SHAP supports both local explanations and global analyses through aggregated feature importance, commonly visualized using informative plots (Lundberg et al., 2020). TreeSHAP (Lundberg et al., 2020) provides exact and efficient computation for tree-based models. A key property for our setting is that SHAP can explain any deterministic function of the model inputs (Lundberg and Lee, 2017), not only predicted probabilities, which enables its direct application to conformal outputs such as conformal p-values or reliability metrics (Johansson et al., 2025).

Recent work has begun exploring the intersection of conformal prediction and explainable machine learning, with two distinct objectives: quantifying uncertainty in explanations, and explaining conformal outputs themselves. Our work falls in the second category.

On the first objective, Alkhatib et al. (2022) proposed using Venn prediction to assess the quality of rule-based explanations, and Alkhatib et al. (2023) explored using conformal regression to approximate SHAP values efficiently while providing validity guarantees for the approximations. The Calibrated Explanations framework (Löfström et al., 2024, 2025) integrates Venn-Abers calibration with feature importance, yielding uncertainty quantification for predictions and their explanations. Mehdiyev et al. (2025) combined permutation feature importance with split conformal prediction to study how feature significance relates to model confidence by permuting features in test data, calibration data, or both.

On the second objective, Johansson et al. (2025) introduced, as proof of concept, the use of SHAP analysis for conformal classification, treating the p-values associated with individual class labels as model outputs suitable for Shapley value-based feature attribution. A key finding is that feature effects on p-values need not be symmetrically distributed across classes, enabling class-specific explanations that are not possible with standard probability outputs. They applied this framework on benchmark datasets using a Random Forest with class-conditional calibration, employing SHAP with default settings as implemented in the SHAP Python package (Johansson et al., 2025). Our work extends this direction with an explicit computational route: the conformal p-value composes the classifier output with a calibration-dependent step function external to the model, which places it outside the reach of tree-specific algorithms such as TreeSHAP; we therefore apply KernelSHAP to the full conformal pipeline, including the empirical calibration step, so that the explained functions are the actual p-values and the margins derived from them. Yapicioglu et al. (2024) proposed ConformalSight, a global explainability framework that measures feature importance by perturbing features in the calibration set and observing changes in aggregate conformal metrics such as coverage and set size. Idrissi et al. (2025) attributed predictive uncertainty to input features by defining cooperative games where conformal interval properties serve as value functions, using Shapley values and Harsanyi allocations in a regression setting.

We extend the second line of work. First, we propose FCOD plots as a population-level diagnostic that reveals where in the feature space the conformal predictor is reliable or uncertain. Secondly, we introduce conformal p-value margins as a SHAP target that captures prediction reliability rather than class-specific evidence.

Unlike ConformalSight, which produces a single global ranking per dataset, our approach provides instance-level attributions through SHAP that can also be aggregated globally. And in contrast to Idrissi et al. (2025), who define custom cooperative games for regres-

sion intervals, we apply KernelSHAP as a model-agnostic estimator to the full conformal classification pipeline.

### 3. Methodology

In this section, we first review Mondrian conformal prediction, focusing on conformal p-values and their induced outcome categories. We then present FCOD plots linking these categories to feature values. Next, we introduce reliability metrics derived from p-values, and show how they combine with SHAP values for explainability.

#### 3.1. Mondrian Conformal Prediction

We employ inductive conformal prediction with class-conditional (Mondrian) partitioning (Vovk et al., 2005). The labeled data is split into a proper training set  $\mathcal{Z}^t$  for fitting the underlying classifier  $h$  and a calibration set  $\mathcal{Z}^c$  for computing p-values.

For a test instance  $x_{k+1}$  with candidate label  $\tilde{y} \in \{0, \dots, K-1\}$ , the nonconformity score using the Least Ambiguous set-valued Classifier (LAC) (Sadinle et al., 2019) is:

$$s_{k+1}^{\tilde{y}} = 1 - \hat{P}_h(\tilde{y} \mid x_{k+1}) \quad (1)$$

where  $\hat{P}_h(\tilde{y} \mid x_{k+1})$  is the classifier’s predicted probability for class  $\tilde{y}$ . Then, the Mondrian p-value is computed using only the calibration instances sharing the candidate label:

$$p_{k+1}^{\tilde{y}} = \frac{\left| \{z_i \in \mathcal{Z}^{\tilde{y}} : s_i > s_{k+1}^{\tilde{y}}\} \right| + \theta_{k+1} \left( \left| \{z_i \in \mathcal{Z}^{\tilde{y}} : s_i = s_{k+1}^{\tilde{y}}\} \right| + 1 \right)}{|\mathcal{Z}^{\tilde{y}}| + 1} \quad (2)$$

where  $\mathcal{Z}^{\tilde{y}} \subseteq \mathcal{Z}^c$  is the subset of calibration instances with true label  $\tilde{y}$ , and  $\theta_{k+1} \sim U[0, 1]$  is a random tie-breaking term. This class-conditional formulation ensures that the coverage guarantee  $P(y_{k+1} \in \Gamma_{k+1}^\alpha) \geq 1 - \alpha$  holds within each class separately (Vovk et al., 2005). At significance level  $\alpha$ , the prediction set consists of all labels not rejected:

$$\Gamma_{k+1}^\alpha = \left\{ \tilde{y} : p_{k+1}^{\tilde{y}} \geq \alpha \right\} \quad (3)$$

To analyze conformal classifier behavior, we categorize each prediction based on the prediction set  $\Gamma_{k+1}^\alpha$  and the true label  $y_{k+1}$ . The general categories are:

- **Single Correct (SC):**  $|\Gamma_{k+1}^\alpha| = 1$  and  $y_{k+1} \in \Gamma_{k+1}^\alpha$ . The predictor outputs a single class that matches the true label.
- **Single Incorrect (SI):**  $|\Gamma_{k+1}^\alpha| = 1$  and  $y_{k+1} \notin \Gamma_{k+1}^\alpha$ . The predictor outputs a single class that does not match the true label. Under exchangeability,  $\Pr(\text{SI}) + \Pr(\text{MU}) + \Pr(\text{E}) \leq \alpha$  holds for standard (marginal) conformal prediction (Vovk et al., 2005). Under Mondrian conformal prediction, the analogous guarantee is class-conditional:  $\Pr(\text{SI} \mid y = k) \leq \alpha$  for each class  $k$  separately, whereas the marginal SI rate need not satisfy this bound exactly (in binary classification,  $\Pr(\text{MU}) = 0$  by construction).
- **Multi-Set (MS):**  $|\Gamma_{k+1}^\alpha| > 1$ . The predictor cannot reject multiple classes at significance level  $\alpha$ , reflecting uncertainty. It becomes Two-Set (TS) in binary settings.

- **Multiset Covered (MC) and Multiset Uncovered (MU):**  $y_{k+1} \in \Gamma_{k+1}^\alpha$  if the true label is among the retained classes, and  $y_{k+1} \notin \Gamma_{k+1}^\alpha$  otherwise.
- **Empty (E):**  $|\Gamma_{k+1}^\alpha| = 0$ . All classes are rejected. This case is rare at typical significance levels and is excluded from the analysis.

For binary classification, the MS (or TS) category always produces the full set  $\{0, 1\}$ . In some cases, we refine MS into two subcategories based on the true label: **TS0** when  $y_{k+1} = 0$  and **TS1** when  $y_{k+1} = 1$ . This distinction is useful because the conformal predictor may show different uncertainty patterns depending across true classes. Thus, separating them using FCOD plots reveals class-specific reliability regions.

### 3.2. Feature-Conditioned Outcome Distribution Plots

As discussed above, conformal prediction assigns each test instance to a category from a finite set of mutually exclusive outcome categories determined by the structure of the prediction set. These outcomes satisfy  $P(\text{SC}) + P(\text{SI}) + P(\text{TS}) + P(\text{E}) = 1$  and, by the marginal coverage guarantee,  $P(\text{SI}) + P(\text{MU}) + P(\text{E}) \leq \alpha$ .

The Feature-Conditioned Outcome Distribution (FCOD) plots visualize how the conditional empirical distribution of these outcomes varies as a function of the values of feature  $X_j$ , thereby linking regions of the feature space to outcome categories that characterize the predictor’s confidence and correctness. Although we focus on numerical features in what follows, the extension to categorical variables is straightforward, since the outcome distributions can be estimated separately for each category level.

The process described below relies on a set of pragmatic parameter choices that we adopt because they produce stable and interpretable plots in the cases analyzed. However, alternative configurations could be considered. A more systematic sensitivity analysis of these design decisions is valuable but lies outside the scope of this paper, whose focus is on evidencing the underlying predictive behavior rather than on the visualization tool itself.

**Step 1: Grid construction.** For each feature  $X_j$ , a uniform evaluation grid  $\mathcal{G}_j = \{g_1, \dots, g_M\}$  consisting of  $M$  points is constructed over the observed range of the feature. For numerical features with approximately symmetric or Gaussian-like distributions, choosing  $M = 50$  and restricting the grid to an interquantile range such as  $[Q_5(X_j), Q_{95}(X_j)]$  is generally adequate. However, features exhibiting more irregular distributions may require tailored grid construction strategies.

**Step 2: Binned proportion estimation.** To estimate the conditional outcome probabilities, we construct a finer auxiliary grid over the same feature range, consisting of  $M_{\text{fine}}$  equally spaced bins. Each instance  $i$  is then assigned to its nearest bin, denoted by  $k^*(i)$ . For each outcome category  $c \in \{\text{SC}, \text{SI}, \text{TS}\}$  ( $\{\text{SC}, \text{SI}, \text{MC}, \text{MU}\}$  in the multiclass) and each bin  $k$ , we compute the empirical proportion

$$\tilde{p}_k^{(c)} = \frac{n_k^{(c)}}{n_k}, \quad (4)$$

where  $n_k$  denotes the number of instances assigned to bin  $k$ , and  $n_k^{(c)}$  denotes the number of those instances associated with outcome category  $c$ . Bins with  $n_k = 0$  are assigned  $\tilde{p}_k^{(c)} = 0$  prior to the smoothing step.

In our implementation, we use the heuristic choice  $M_{\text{fine}} = \max(2M, 200)$ , which yielded stable estimates across the dataset considered in this work, although alternative resolutions could also be adopted depending on the application and sample size.

**Step 3: Gaussian smoothing and renormalization.** To obtain smoother estimates of the conditional outcome distributions, a smoothing operator is applied independently to each outcome probability curve.

$$\hat{p}_k^{(c)} = \sum_{k'} \tilde{p}_{k'}^{(c)} \phi(k - k'; \sigma), \quad (5)$$

where  $\phi(\cdot; \sigma)$  denotes a discrete smoothing kernel parameterized by a bandwidth  $\sigma$ .

We use a discrete Gaussian kernel with  $\sigma = 2.0$  grid units, which provides a suitable trade-off between smoothness and local fidelity in the experiments considered. Since this approach may lead to over-smoothing in high-density regions and under-smoothing in low-density ones, it could be further refined by adopting a data-adaptive bandwidth.

Since each outcome category is smoothed independently, the resulting curves generally do not preserve the probability simplex constraint. To restore a valid probability distribution at each grid point, the smoothed values are re-normalized point-wise:

$$\bar{p}_k^{(c)} = \frac{\hat{p}_k^{(c)}}{\sum_{c'} \hat{p}_k^{(c')}}, \quad (6)$$

thereby ensuring that  $\sum_c \bar{p}_k^{(c)} = 1$  for every grid point  $k$ . Grid points with zero total smoothed mass are assigned the uniform distribution  $1/|\mathcal{C}|$  as a fallback placeholder.

Finally, the resulting fine-grid curves are down-sampled to the target resolution of  $M$  evaluation points. In our implementation, this is performed through linear interpolation followed by a second application of Equation (6).

**Step 4: Stacked area visualization.** The smoothed conditional outcome distributions are represented as vertically stacked areas with cumulative boundaries defined by:

$$b_k^{(0)} = 0, \quad b_k^{(r)} = \sum_{r' \leq r} \bar{p}_k^{(c_{r'})}, \quad r = 1, \dots, |\mathcal{C}|, \quad (7)$$

where  $b_k^{(r)}$  denotes the cumulative stacked height up to and including the  $r$ -th outcome category at grid point  $k$ ; the outcome categories are ordered as SC, SI, and TS (or SC, SI, MC, MU in the multiclass case), so the stacked representation sums to one at every evaluation point  $g_k$ .

To provide information about the distribution of samples across the feature domain, a marginal histogram of  $X_j$  computed over the test set is displayed below the main panel. This visualization facilitates the interpretation of the estimated outcome distributions by indicating regions with both high and low sample density.

**Interpretation of the FCOD plots.** FCOD plots provide a global diagnostic of conformal predictor behavior by visualizing the empirical conditional distribution of discrete outcome categories as a function of a single feature  $X_j$ . Each plot conditions on  $X_j$  while implicitly marginalizing over the empirical joint distribution of all remaining features. As

a result, FCOD curves reflect not only the observed association with  $X_j$ , but also any structure induced by its dependence with other predictors.

This marginalization implies that FCOD profiles are inherently associative rather than causal: when  $X_j$  is correlated with other covariates, its apparent effect may partially reflect the influence of confounding variables. Consequently, similar FCOD shapes across features do not necessarily imply similar roles within the conformal classifier.

FCOD plots are therefore best understood as descriptive tools that localize regions of the feature space associated with reliable, unreliable, or ambiguous prediction behavior. However, they do not disentangle the individual contribution of input features. To address this limitation, we complement FCOD analysis with SHAP-based explanations in Section 3.4. While FCOD plots answer *where* in the feature space reliability or uncertainty concentrates, SHAP analysis of the correctness margin provides a complementary *why*-level explanation at the instance level, attributing prediction reliability to individual input features.

### 3.3. Conformal Reliability Metrics

For a test instance  $x_{k+1}$ , Mondrian conformal prediction provides class-conditional p-values for each candidate label  $\tilde{y} \in \{0, \dots, K-1\}$ , as defined in Eq. 2. These p-values quantify the compatibility of the instance with each class under the fitted conformal model. By aggregating them across all candidate labels, we can define summary measures that characterize the conformal reliability of the prediction; when the ground-truth label is available, these measures further allow us to assess whether the prediction is ultimately correct or erroneous.

**Evidence margin.** Let  $p^i(x)$  and  $p^j(x)$  denote the conformal p-values for the two classes  $i$  and  $j$ , the pairwise evidence margin can be defined as:

$$\delta_{\text{em}}^{i,j}(x) = p^i(x) - p^j(x) \quad (8)$$

This quantity is defined independently of the true label and measures the degree of separation between competing hypotheses.

For binary classification, it measures the signed difference in conformal evidence between the two classes and values close to zero indicate ambiguity. However, in a multiclass setting, in practice it is often most informative to interpret it in a one-vs-rest or top-two comparison, where the metric highlights the relative strength of the most competitive alternatives.

**Correctness margin.** Let  $y_{\text{true}}$  denote the true label of a test instance  $x_{k+1}$ , and let  $p^j(x)$ , for  $j \in \{0, \dots, K-1\}$ , denote its class-conditional conformal p-values. We define the *correctness margin* as:

$$\delta_{\text{cm}}(x) = p^{y_{\text{true}}}(x) - \max_{y \neq y_{\text{true}}} p^y(x), \quad (9)$$

This quantity lies in  $[-1, 1]$  and summarizes the extent to which conformal evidence favors the true class over its strongest competing alternative.

A large positive value indicates that the true class clearly dominates, corresponding to a confident and correct singleton prediction. A negative value indicates that at least one incorrect class carries stronger conformal evidence than the true class. Values close to zero reflect similar evidence across classes, typically leading to non-singleton prediction sets (either multi-set or empty) depending on the chosen significance level  $\alpha$ .

The value  $\delta_{\text{cm}}(x)$  is a label-dependent measure, as it requires knowledge of the ground-truth label and cannot be computed at inference time. It is used as a diagnostic tool to evaluate the correctness of conformal predictions and their associated confidence patterns.

### 3.4. SHAP Decomposition of the Margins

Building upon the SHAP-based analysis of conformal p-values proposed by [Johansson et al. \(2025\)](#), we use the reliability metrics introduced in the previous section as SHAP targets. Whereas SHAP decompositions of individual p-values explain class-specific conformal evidence, our approach focuses instead on explaining prediction reliability through the evidence margin and correctness margin, each treated as a target function for feature attribution. This yields a feature-level decomposition of both class separation and prediction correctness.

We use SHAP to decompose the proposed margins into additive feature contributions. SHAP is not introduced here as a new explanation algorithm; rather, it operationalizes the proposed conformal targets. For the evidence margin, the resulting attributions identify which features shift relative conformal evidence toward one class or the other. For the label-alignment margin, they identify which features strengthen or weaken the relative evidence supporting the observed class. The scientific contribution therefore lies in the definition and interpretation of the explained quantities and in their integration with FCOD, rather than in the generic application of SHAP itself.

For a test instance  $x$  with feature vector  $(x_1, \dots, x_d)$ , SHAP decomposes a given margin-based target into additive feature contributions  $\phi_1, \dots, \phi_d$  such that:

$$\delta(x) = \phi_0 + \sum_{j=1}^d \phi_j \quad (10)$$

where  $\phi_0 = \mathbb{E}[\delta(x)]$  denotes the expected margin over the reference dataset and  $\phi_j$  represents the contribution of feature  $j$  to the margin associated with the instance  $x$ .

Features with positive SHAP values increase the target margin relative to its expected value, thereby strengthening either the separation between competing classes or the support for the true class, depending on the margin under consideration. Conversely, features with negative SHAP values decrease the margin, reducing class separation or weakening the evidence in favor of the true class.

The linearity of Shapley values ([Shapley, 1953](#)) provides a direct link between margin SHAP and per-class p-value SHAP. For the correctness margin:

$$\phi_j^{(\delta_{\text{cm}})} = \phi_j^{(y_{\text{true}})} - \phi_j^{(y^*)} \quad (11)$$

where  $y^* = \arg \max_{y \neq y_{\text{true}}} p^y(x)$  is the most competitive incorrect class and  $\phi_j^{(y)}$  denotes the SHAP value of feature  $j$  for the p-value of class  $y$ . In our implementation, KernelSHAP is applied once to the p-value function  $[p^0(x), p^1(x)]$ , and margin SHAP values are obtained by algebraic subtraction of the resulting per-class attributions, as in Eq. (11). This procedure is exact for binary classification, where  $y^*$  is always the unique alternative class, so the subtraction involves fixed coefficients and Shapley linearity applies without qualification.

For a binary classification problem:

$$\phi_j^{(\delta_{\text{em}})} = \phi_j^{(1)} - \phi_j^{(0)}, \quad (12)$$

where  $\delta_{\text{em}}$  is the evidence margin for a binary classification problem as defined in Eq.(8).

A feature may increase either margin by increasing the conformal p-value associated with the favored class, by decreasing that of the competing class, or through a combination of both effects. In the multiclass setting, the competing class  $y^*$  may vary across instances, implying that global aggregations of  $\phi_j^{(\delta_{\text{cm}})}$ , such as beeswarm plots, combine explanations relative to different competing classes. By contrast, the binary evidence margin  $\phi_j^{(\delta_{\text{em}})}$  is defined with respect to a fixed pair of classes and therefore admits a more direct global interpretation. For this reason, it constitutes the primary SHAP target in the binary experiments reported in Section 4.

## 4. Experiment Setup

In this section we will present the experiments carried out, the results of which we will show in Section 5. The code and data needed to run them are available on GitHub<sup>2</sup>.

### 4.1. Synthetic dataset

To demonstrate that the proposed tools generalize beyond binary classification, we construct a synthetic 3-class benchmark of 12,000 instances where class priors are imbalanced (0.5/0.3/0.2). The instances are described by 8 features: six of them are noise with no effect on the dataset construction, while the other two features ( $x_0, x_1$ ) are used to create three Gaussian clusters with standard deviation ( $\sigma = 1.3$ ), calibrated to yield approximately 20% multi-set predictions at  $\alpha = 0.10$ . The cluster centers are the vertices of an equilateral triangle in  $(x_0, x_1)$ , keeping the three pairwise class boundaries symmetric so that the  $(y, y^*)$  comparisons are performed in comparable terms (Figure 1). The dataset is split into training, calibration, and test sets using a 60/20/20 stratified partition.

We describe next the model configuration. Since the target variable is imbalanced, to better account for the conditional coverage of the minority class, we use Mondrian conformal prediction (Boström et al., 2021) at  $\alpha = 0.10$ . We train an XGBoost classifier (Chen and Guestrin, 2016) on the training set. The calibration set is used to compute Mondrian conformal p-values, and the test set is used for all reported evaluations. We apply Mondrian inductive conformal prediction with Least Ambiguous set-valued Classifier (LAC) nonconformity scores (Sadinle et al., 2019), as described in Section 3.1, at significance level  $\alpha = 0.10$ . The Mondrian partitioning ensures class-conditional coverage.

We also use KernelSHAP with `nsamples=1024`, which exceeds  $2^8 - 2 = 254$ , so the attributions are exact with zero sampling variance. FOD plots are computed over a grid of 50 equally-spaced points with a Gaussian smoothing bandwidth  $\sigma = 2.0$ .

### 4.2. Hotel Booking Demand dataset

As a real-world case study, we use the Hotel Booking Demand dataset (Antonio et al., 2019), released as a research resource and also available on Kaggle<sup>3</sup>. We focus on the Lisbon city hotel subset (H2). Each observation represents a hotel booking and is characterized by 31 variables taken from the hotel’s PMS (Property Management System).

2. <https://github.com/Caparrini/phd-conformal-experiments/tree/master/experiments/copa2026>

3. <https://www.kaggle.com/datasets/mojtaba142/hotel-booking>

After stratified random splitting (60% training, 20% calibration, 20% test), the working sample comprises 79,330 instances with a stable cancellation rate of 41.7% across all splits. Although the original dataset contains additional variables, we restrict the analysis to a subset of readily interpretable features, four numerical and four categorical as shown in Table 1. This controlled set facilitates clearer visualization and analysis of the explainability tools and their connection to conformal prediction outcomes.

| Variable                 | Type        | Description                                                  |
|--------------------------|-------------|--------------------------------------------------------------|
| lead_time                | Numerical   | Number of days between booking date and arrival date.        |
| arrival_date_week_number | Numerical   | Week number of the arrival date.                             |
| stays_total_nights       | Numerical   | Total number of nights booked.                               |
| adr                      | Numerical   | Average daily rate of the booking.                           |
| market_segment           | Categorical | Market segment in 7 categories.                              |
| customer_type            | Categorical | Booking type (Contract, Group, Transient, Transient-party).  |
| deposit_type             | Categorical | Deposit status (no deposit, non refund, refundable).         |
| is_repeated_guest        | Binary      | Indicates whether the guest is a repeated customer.          |
| is_canceled              | Binary      | Target variable indicating whether the booking was canceled. |

Table 1: Description of the variables used in the Hotel Booking Demand dataset

The procedure followed is analogous to the one we used in the synthetic dataset. Regarding model configuration, we apply Mondrian inductive conformal prediction with Least Ambiguous set-valued Classifier (LAC) nonconformity scores (Sadinle et al., 2019), as described in Section 3.1, at significance level  $\alpha = 0.10$ . In this case, the XGBoost model hyperparameters are: 200 estimators, maximum depth 6, learning rate 0.1, subsample ratio 0.8, and column subsampling ratio 0.8, with a fixed random seed 42.

KernelSHAP is applied to the conformal p-value function, returning per-class SHAP attributions. Margin SHAP (see Eq. 12) values are then derived by algebraic subtraction, exploiting Shapley linearity. We use a background dataset of 300 instances randomly sampled from the test set, we explain 1000 instances, and use 256 perturbation samples per instance (`nsamples=256`), with the identity link function. Since  $256 \geq 2^8 - 2$ , KernelSHAP enumerates the full coalition lattice over the  $d = 8$  features, so the resulting attributions are exact Shapley values not sampling-based approximations. FOD plots are computed over a grid of 50 equally-spaced points with a Gaussian smoothing bandwidth  $\sigma = 2.0$ .

## 5. Results

### 5.1. Synthetic dataset

Figure 1 shows the conformal outcomes in the  $(x_0, x_1)$  space and Table 2 reports the classifier and conformal metrics. The XGBoost classifier recovers the designed class structure, the conformal approach delineates the reliability regions, while Mondrian coverage matches the  $1 - \alpha = 0.90$  target across all three classes despite the 0.5/0.3/0.2 prior imbalance.

Because the generative process is known, exact Shapley values can be computed by enumerating all coalitions under the oracle Bayes posterior (GT-Shapley). Against this reference, KernelSHAP achieves 93.3% sign agreement on the informative features and an informative-to-noise attribution ratio between 12 and 14 times, whereas noise-feature attributions are exactly zero. It demonstrates that our approach reliably approximates the true feature contributions.

Table 2: Synthetic 3-class benchmark: model, conformal, and fidelity metrics.

| Category     | Metric                       | Value             | Description                  |
|--------------|------------------------------|-------------------|------------------------------|
| Model        | Test Accuracy                | 0.845             | vs. Bayes-optimal 0.854      |
|              | Balanced Accuracy            | 0.823             | Macro-averaged recall        |
|              | Recall (Class 0/1/2)         | 0.899/0.812/0.759 | Prior 0.5/0.3/0.2            |
|              | Precision (Class 0/1/2)      | 0.867/0.817/0.830 |                              |
|              | ROC-AUC (Macro, One vs Rest) | 0.954             |                              |
|              | ROC-AUC (Macro, One vs One)  | 0.953             |                              |
| Coverage     | Marginal                     | 0.914             | Target: $1 - \alpha = 0.90$  |
|              | Class 0                      | 0.908             | $n = 1202$                   |
|              | Class 1                      | 0.919             | $n = 729$                    |
|              | Class 2                      | 0.919             | $n = 469$                    |
| Outcome Rate | SC                           | 0.708             | Single Correct               |
|              | SI                           | 0.072             | Single Incorrect             |
|              | MC                           | 0.205             | Multi-set Covered            |
|              | MU                           | 0.014             | Multi-set Uncovered          |
|              | Empty                        | 0.000             |                              |
| Efficiency   | Multi-set Rate (MC+MU)       | 0.219             | $\approx 20\%$ design target |
|              | Avg. Set Size, Marginal      | 1.234             |                              |
|              | Avg. Set Size (Class 0/1/2)  | 1.202/1.265/1.267 |                              |
| Fidelity     | GT-Shapley Pearson           | 0.909/0.986       | Mean/median vs. oracle       |
|              | Sign Agreement (Informative) | 0.933             | vs. oracle Shapley           |
|              | Informative/Noise SHAP Ratio | $13.7\times$      | Oracle: 0 on noise           |

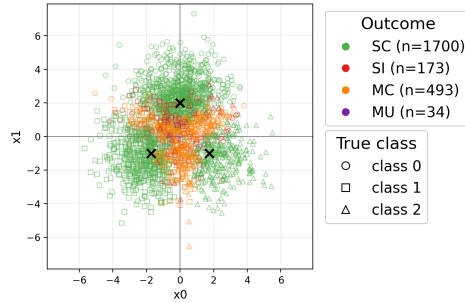
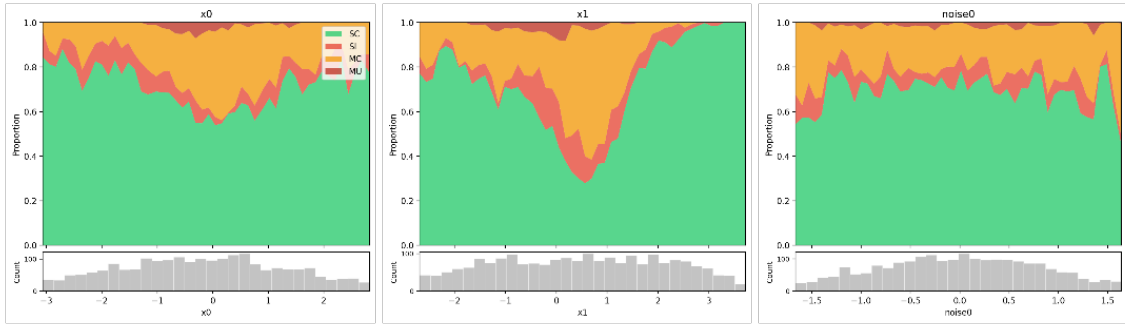


Figure 1: Conformal outcomes in the synthetic 3-class dataset (cluster centers marked).

Figure 2: FCOD plots for the synthetic 3-class benchmark ( $x_0$ ,  $x_1$ , and  $noise0$ ).

The FCOD plots in Figure 2 confirm that outcome proportions vary sharply across the informative features  $(x_0, x_1)$ , concentrating MC and MU instances near class boundaries. Lower uncertainty is observed for high values of  $x_1$ , as in our dataset those values correspond exclusively to class-0 instances. In contrast, the six noise features remain flat (only one of them is shown in the plot) and the largest observed deviation from the marginal rate (0.096) matches the value expected under pure sampling noise ( $0.097 = \hat{\sigma} \sqrt{2 \ln(240)}$ , the extreme-value approximation for the maximum absolute deviation across  $6 \times 10 \times 4 = 240$  bin-outcome comparisons on the noise features, with  $\hat{\sigma}$  the worst-case per-bin binomial standard error).

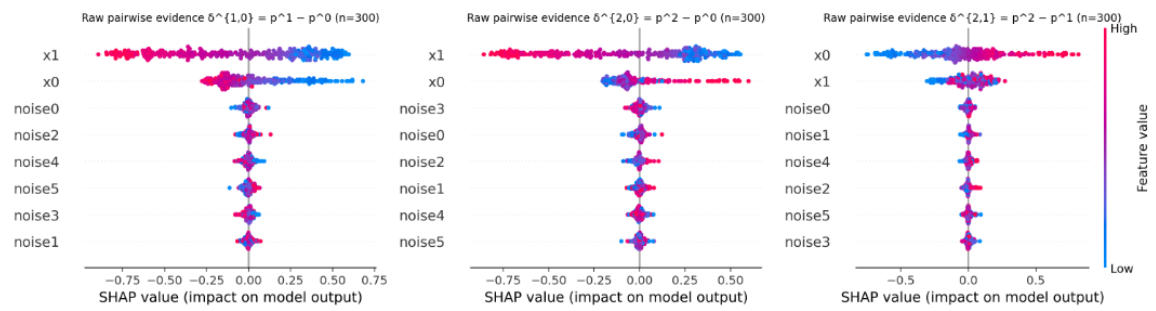


Figure 3: SHAP beeswarms of the pairwise conformal p-value differences  $\delta^{1,0}$ ,  $\delta^{2,0}$  and  $\delta^{2,1}$ .

Finally, we compute the evidence margin of the pairwise comparisons of the three classes ( $\delta^{1,0} = p^1 - p^0$ ,  $\delta^{2,0} = p^2 - p^0$ ,  $\delta^{2,1} = p^2 - p^1$ ) as shown in Figure 3. According to the SHAP beeswarm plots, the SHAP values correctly rank  $x_0$  and  $x_1$  above all six noise features, reflecting how the algorithm uses high or low values of these variables to discriminate among the three classes in a manner consistent with the dataset’s structure.

## 5.2. Hotel Booking Demand dataset

The classifier performance is assessed using standard classification metrics (ROC-AUC, balanced accuracy, F1, precision and recall) across the three splits (Train, Calibration, and Test). Table 3 shows consistent performance across splits, with all metrics falling within the 0.75–0.90 range, indicating stable and adequate predictive performance for the purposes of conformal prediction and subsequent explainability analysis.

Table 3: Machine learning classification performance metrics across splits.

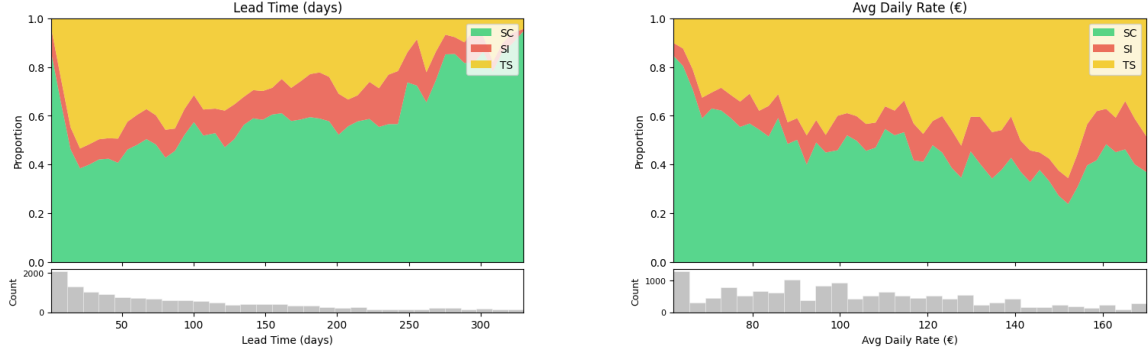
| Metric            | Train | Calibration | Test  |
|-------------------|-------|-------------|-------|
| ROC-AUC           | 0.898 | 0.871       | 0.863 |
| Balanced Accuracy | 0.787 | 0.764       | 0.752 |
| F1 (macro)        | 0.797 | 0.772       | 0.759 |
| Precision (macro) | 0.841 | 0.814       | 0.804 |
| Recall (macro)    | 0.787 | 0.764       | 0.752 |

Table 4 reports the conformal evaluation metrics on the test set, confirming empirical coverage properties at  $\alpha = 0.10$ . Since the dataset shows a singleton rate of 69.7% (of which 59.5% correspond to correct singleton prediction) and includes multi-set predictions, which makes it suitable to evaluate both FCOD plots and SHAP margin analysis. These tools

Table 4: Mondrian Inductive Conformal Prediction results at significance level  $\alpha = 0.10$ .

| Category   | Metric               | Value  | Description                                                           |
|------------|----------------------|--------|-----------------------------------------------------------------------|
| Coverage   | Total Coverage       | 0.897  | Target: $1 - \alpha = 0.90$                                           |
|            | Coverage (Class 0)   | 0.896  | Non-cancelled bookings                                                |
|            | Coverage (Class 1)   | 0.900  | Cancelled bookings                                                    |
|            | $\hat{q}$ (Class 0)  | 0.468  | Nonconformity threshold, class 0                                      |
|            | $\hat{q}$ (Class 1)  | 0.761  | Nonconformity threshold, class 1                                      |
| Efficiency | Singleton Rate       | 0.697  | $ \Gamma^\alpha  = 1$                                                 |
|            | SC Rate              | 0.595  | Singleton correct ( $ \Gamma^\alpha  = 1, y \in \Gamma^\alpha$ )      |
|            | SI Rate              | 0.103  | Singleton incorrect ( $ \Gamma^\alpha  = 1, y \notin \Gamma^\alpha$ ) |
|            | Uncertainty Rate     | 0.303  | $ \Gamma^\alpha  = 2$ , i.e. $\{0, 1\}$                               |
|            | Empty Rate           | 0.000  | $ \Gamma^\alpha  = 0$                                                 |
|            | Avg. Set Size        | 1.303  | Mean prediction set cardinality                                       |
| Sample     | Calibration/Test $n$ | 15,866 | Instances per split                                                   |

allow us to interpret the relationship between feature ranges and the model’s uncertainty and its propensity for misclassification. For the sake of brevity, we will not analyze all variables, but rather the most relevant and illustrative ones.

Figure 4: FCOD plots for the numerical variables `lead_time` and `adr`

### 5.3. Analysis of the feature-conditioned outcome distributions

In this section, we will analyze the relationship between the features and the outcome distribution. Figure 4 shows the FCOD plots for two of the four numerical features. They illustrate how particular ranges align with changes in predictive confidence and correctness, without implying causality. The histogram at the bottom of each plot helps to determine whether the evidence is supported by a large or small number of instances.

The FCOD of `lead_time` reveals that both maximum efficiency and predictive certainty are concentrated at the feature’s extremes, as reflected by the high prevalence of SC outcomes and the low prevalence of SI and TS outcomes. At the lower end of the range, conformal efficiency drops sharply, reaching minimum values between 20 and 30 days before arrival, indicating that the classifier more frequently produces multi-set predictions (TS) and occasional incorrect singleton predictions (SI). At the upper end of the range, efficiency and predictive certainty rise again, but the evidence is much weaker due to the

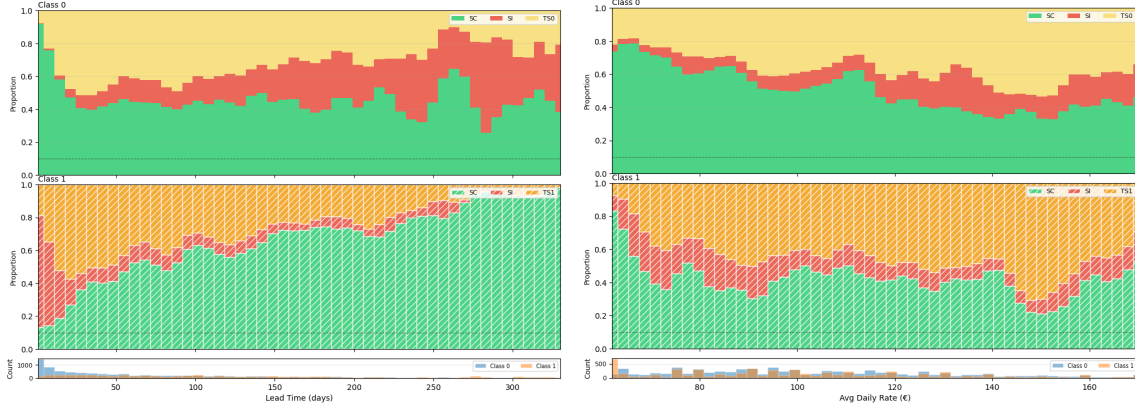


Figure 5: FCOD of `lead_time` (left) and `adr` (right) for class 0 (top) and class 1 (bottom).

small number of instances in this region. Using Figure 5, we can see how these patterns distribute across each class (`is_canceled` = 0 and 1). For class 0, SI outcomes are almost absent at the beginning of the feature range and highly prevalent for class 1. In contrast, this pattern is reversed over the remainder of the range.

For the `adr` variable, we observe that the model exhibits a very high proportion of SC outcomes and a low incidence of both SI and TS outcomes in the lowest range, below 50 euros, indicating a regime of high conformal efficiency and well-calibrated singleton predictions, which may correspond to free stays or heavily discounted bookings. As the range increases, we observe higher uncertainty, as reflected by a higher prevalence of TS outcomes and a more or less constant presence of SI predictions. For the highest values, above 150 euros, SC outcomes increase again but so do SI predictions, suggesting a more heterogeneous conformal regime. However, it should be noted that the upper range is supported by a substantially lower number of observations, which warrants caution when interpreting the observed conformal behavior in this region.

Figure 6 presents the FCOD plots for the categorical values, showing the distributions of the different conformal prediction outcomes across categories. The number of instances in each category is displayed at the top of each bar. Furthermore, the categories are ordered from highest to lowest according to the proportion of SC outcomes, with higher SC proportions indicating better conformal behavior. We focus our analysis on the variables that contribute most strongly to the conformal margins, i.e., those with the greatest influence in discriminating between classes according to Figure 7.

For the variable `deposit_type`, the category of **Non Refund** bookings yield 100% SC outcomes. This corresponds to a highly separable region of the feature space, where the conformal predictor consistently produces correct singleton prediction sets. The **Refundable** category ( $n=5$ ) shows no SI outcomes, though sample size limits interpretation. In contrast, the **No Deposit** category exhibits lower conformal quality, with approximately 50% SC outcomes, around 10% singleton incorrect predictions (SI), and the remaining cases corresponding to multi-set predictions (TS). In this region of the feature space the conformal predictor exhibits poorer calibration, as reflected in its difficulty to produce reliable singleton predictions. Under this criterion, the **Groups** category of **Market Segment** and

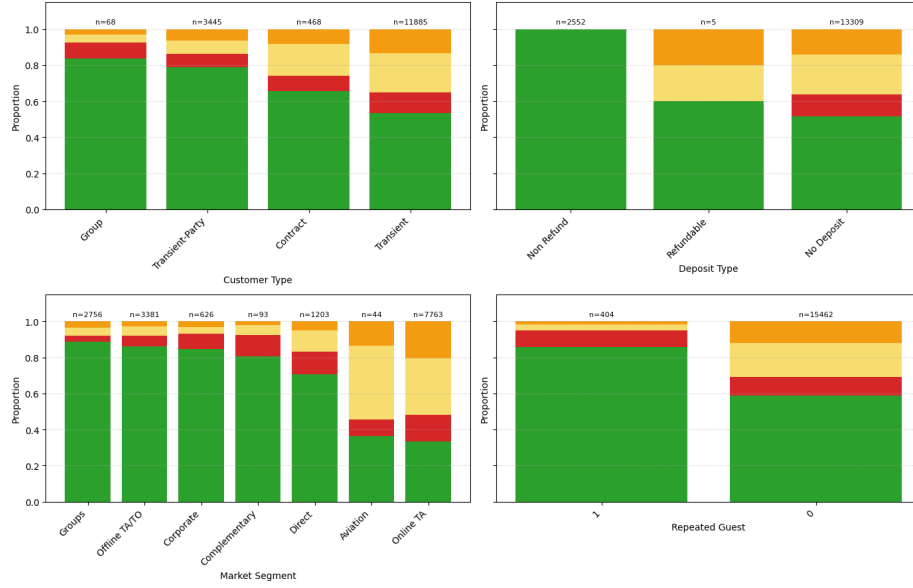
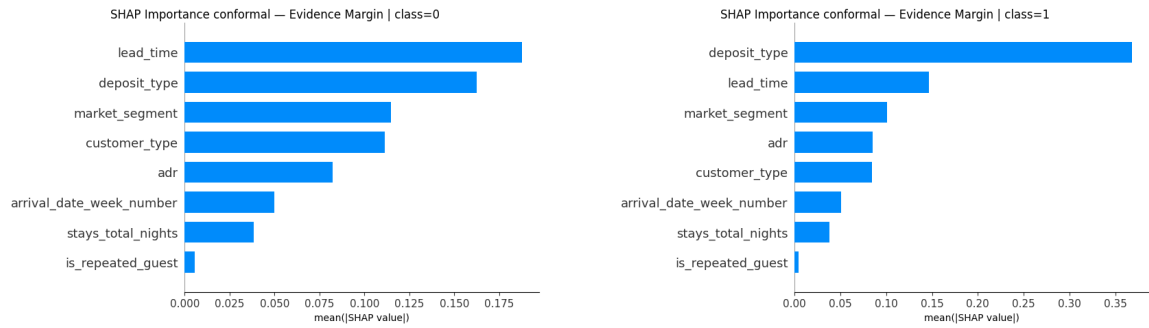


Figure 6: FCOD plots for the categorical features.

the **Group** category of **Customer Type** exhibit the most informative conformal behavior, whereas bookings made through online travel agencies (**Online TA**) and independent bookings (**Transient**) exhibit the least informative patterns, characterized by lower SC rates and higher predictive ambiguity. Finally, in the **Repeated Guest** variable, repeat guests exhibit more reliable and less ambiguous prediction sets than non-repeat guests.

#### 5.4. Explainability analysis of the predictive reliability margins

Figure 7: Feature importance of the conformal evidence margin, measured as  $\text{mean}(|\text{SHAP}|)$  over KernelSHAP explanations. Importance for class 0 (left) and 1 (right).

We compute the evidence and correctness margins and treat each of them as a target function for feature attribution using SHAP. This yields a feature-level decomposition of both class separation and prediction correctness. Figure 7 shows mean absolute SHAP importance for the evidence margin by class, that is, feature importance is computed using only the instances belonging to that class. The resulting figure reveals a marked asymmetry.

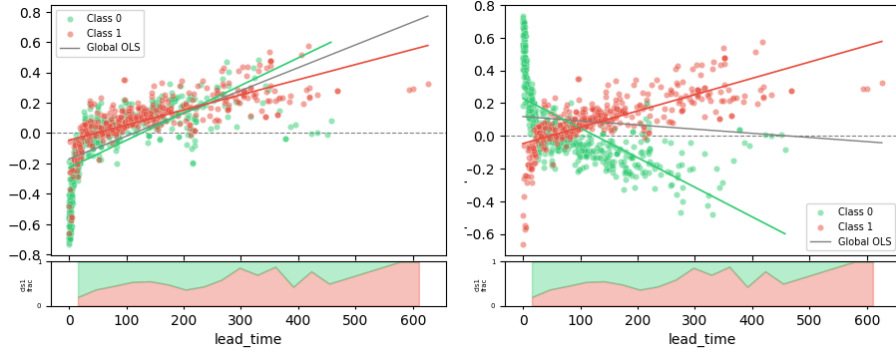


Figure 8: SHAP dependence plots for the conformal correctness (left) and evidence (right) margins for `lead_time`.

For Class 1, `deposit_type` dominates with a mean absolute SHAP roughly 2.5 times that of `lead_time`. For Class 0, the ranking inverts and `lead_time` slightly leads `deposit_type`. In third and fourth place, we find `market_segment` and `customer_type` in both classes. Finally, `is_repeated_guest` shows as the least influential feature in both rankings.

Figure 8 shows SHAP dependence plots for both margins in `lead_time`, which is the most influential numerical feature. It is important to bear in mind that SHAP values represent each feature’s contribution relative to the baseline expectation; a positive value does not by itself imply a positive final margin. For the correctness margin (left panel), `lead_time` shows a monotonically positive SHAP contribution across both classes: longer booking horizons shift conformal evidence toward Class 1 regardless of the true label. If we look at the class-conditional OLS trends, Class 1 exhibits a steeper slope, confirming lead time as the primary driver of the classifier’s directional bias. For the evidence margin (right panel), `lead_time` exhibits a monotone class asymmetry: Class 1 instances accumulate strongly positive SHAP values as lead time increases, corresponding to reliably correct cancellation predictions, while Class 0 instances acquire correspondingly negative margin at the same values. This divergence is increasingly pronounced at long booking horizons, indicating that lead time reinforces singleton-correct predictions for cancellations and degrades reliability for non-cancellations.

Figure 9 shows the dependence plots for `deposit_type`, the most influential categorical feature. The `Non Refund` category pushes both the correctness and evidence margins strongly positive for Class 1, reflecting a strong association between non-refundable bookings and cancellation. Notably, there are no instances with a non-refund deposit that were not canceled. For the `No Deposit` category, the correctness margin shows an overlap between Class 0 and Class 1, while in the evidence margin both classes exhibit similar strength but opposite signs (positive for Class 0, negative for Class 1). This indicates that this category introduces directional ambiguity that varies by true class.

## 6. Conclusions

We introduce a diagnostic framework for conformal classifiers comprising FCOD plots and two p-value-based SHAP targets: the label-free evidence margin and the label-dependent

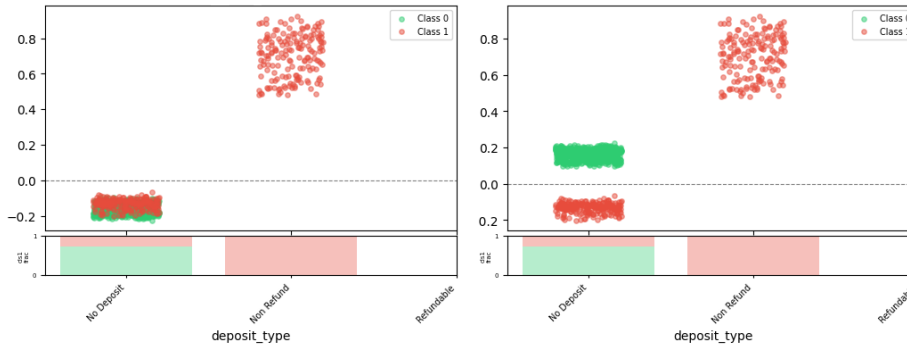


Figure 9: SHAP dependence plots for the conformal correctness (left) and evidence (right) margins for `deposit_type`.

correctness margin. FCOD plots identify where along specific features the conformal predictor shifts across regions of reliability and correctness, without attributing causality. Margin SHAP identifies which and how features drive prediction reliability at the instance level in terms of class separation and correctness, enabling global rankings via mean absolute SHAP value and local decompositions. Together they form a diagnostic workflow, actionable in applied settings, that shifts focus from aggregate accuracy to region-specific reliability, helping detect where conformal decisions require heightened caution before deployment.

The two margins address complementary diagnostic questions. The label-free evidence margin measures the signed difference between the p-values of competing classes and can therefore be computed without access to the true label. The label-dependent correctness margin compares the p-value of the true class with that of its strongest alternative and is intended for retrospective evaluation. Together with the class-wise p-values on which they are based, these quantities distinguish class-specific conformal evidence, relative class separation, and the extent to which the evidence favors the true class.

The explanatory contribution of these margins does not depend on a non-monotonic relationship between predicted probabilities and conformal p-values. Under LAC and fixed class-conditional calibration, each p-value is a nondecreasing empirical step transformation of the corresponding predicted probability. Nevertheless, probabilities, p-values, and p-value margins remain distinct explanatory targets: probabilities represent classifier outputs, p-values express class-specific compatibility relative to the calibration data, and the margins explicitly contrast the conformal evidence assigned to competing classes.

Applied to both a synthetic and a real-life dataset as a proof of concept, the framework surfaces feature-level reliability structure that aggregated conformal metrics conceal, demonstrating consistent diagnostic signals across both numerical and categorical features.

Further research on integrating conformal prediction and explainability is needed to develop diagnostic tools that characterize both the strengths and failure modes of conformal decision systems for downstream decision-making. Future work should explore domain-specific applications in high-uncertainty real-world settings, extensions to conformal regression, and how feature importance and reliability patterns vary with the significance level. Incorporating these diagnostic tools into decision-making frameworks validated by domain experts is also an important direction for assessing their practical utility.

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